

Towards Human-like Empathy: Confidence-Driven Fast-Slow Reasoning for Multimodal Empathetic Response Generation

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Abstract

Multimodal Empathetic Response Generation (MERG) integrates textual, visual, and acoustic signals to synthesize nuanced affective responses. While Large Language Models (LLMs) demonstrate strong capabilities in emotional reasoning, they still suffer from reasoning redundancy and contextual prolixity. Applying static, heavy-duty reasoning to all queries wastes computational resources, while uncompressed multimodal histories introduce excessive and noisy tokens into LLM inputs. To address these challenges, we propose the CoRE framework. Inspired by dual-process cognition, CoRE introduces a confidence-driven fast-slow reasoning mechanism that dynamically toggles between a lightweight fast path for intuitive fluency and an expanded slow reasoning chain for resolving complex emotional ambiguity. Additionally, a condensed-temporal dialogue encoder module distills noisy historical signals into compact representations, reducing the multimodal token burden. Extensive experiments on the large-scale AvaMERG benchmark demonstrate that CoRE achieves competitive performance, outperforming existing LLM baselines in emotion recognition, semantic alignment and human-evaluated empathy while optimizing inference efficiency. The source code is available at https://github.com/Cathryn1103/CoRE_MERG.

CCS Concepts

• **Computing methodologies** → **Discourse, dialogue and pragmatics**; • **Information systems** → **Multimedia information systems**; **Sentiment analysis**; • **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Empathetic response generation; multimodal dialogue; emotional reasoning; fast-slow thinking; large language models

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1 Introduction

Multimodal Empathetic Response Generation (MERG) [3, 24, 51] extends traditional text-based dialogue [25, 31, 36] by integrating paralinguistic signals, such as speech prosody and facial dynamics [13, 29], to generate empathetic responses. While contemporary Large Language Models (LLMs) [22, 48] exhibit advanced cognitive reasoning, they frequently fall short of achieving human-like empathy. Real-world emotions are rarely confined to explicit semantic labels; rather, they are embedded within subtle, synchronous multimodal streams [5, 35, 51, 58]. Consequently, models relying exclusively on textual modalities may lose critical affective cues, leading to performative or generic responses.

To bridge this sensory gap, recent advancements increasingly leverage deep reasoning mechanisms, such as Chain-of-Thought (CoT) prompting [4, 48] and external knowledge graphs [15, 37], to decode high-level cognitive insights from raw inputs. However, while these works emphasize the interpretability of emotional reasoning, they overlook two key bottlenecks in multimodal empathetic systems as visually summarized in Figure 1:

- (1) **Reasoning Redundancy:** Existing models typically impose a static, heavy-duty reasoning path for every input. This “over-thinking” incurs computational costs for straightforward queries that only require intuitive, fast responses.
- (2) **Contextual Prolixity:** The inherent noise and asynchronous nature of long multimodal dialogue histories often lead to redundant token calculations, overwhelming the LLM and diluting salient affective transitions [40].

To overcome these limitations, we propose **CoRE** (Confidence-driven fast-slow Reasoning for multimodal Empathetic response generation). Inspired by the dual-process theory of human cognition and recent advances in LLM reasoning [22, 27, 33, 39], CoRE



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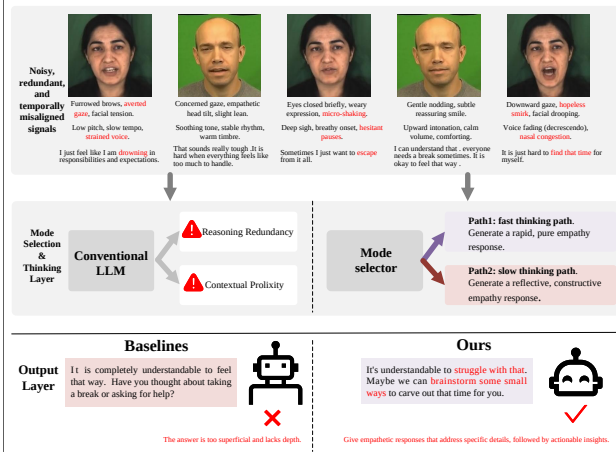


Figure 1: Comparison of generation paradigms. To overcome the reasoning redundancy and contextual prolixity of conventional LLMs, our CoRE routes noisy multimodal signals into fast or slow thinking paths, yielding empathetic responses.

balances inference efficiency and cognitive depth through two primary innovations:

Confidence-Driven Fast-Slow Reasoning. Instead of a static inference pipeline, CoRE dynamically routes the cognitive path based on the model’s confidence (e.g., entropy) [46]. As a parameter-free routing signal, the entropy switches between two modes: for outputs with high certainty (low entropy), the model executes a lightweight *Fast Path* to ensure conversational fluency and minimal latency; when encountering emotional ambiguity (high entropy), it activates an expanded *Slow Reasoning* chain for deeper analysis of latent intents and complex causality. This mechanism ensures that expensive computational power is allocated strictly when necessary.

Condensed-Temporal Dialogue Encoder. To align multimodal inputs with reasoning objectives, we propose the Condensed-Temporal Dialogue (CTD) encoder. This module utilizes reasoning instructions as semantic cues to guide the extraction and fusion of multimodal contexts across temporal sequences. By condensing historical dialogue turns through a multimodal interaction mechanism, the encoder selects emotionally salient information to capture subtle affective changes. This targeted process reduces the semantic gap between different modalities. Consequently, this architecture ensures that visual and acoustic signals are aligned with the LLM’s textual reasoning space, providing an informative representation for complex emotional reasoning tasks.

The main contributions of this paper are summarized as follows:

- We propose CoRE, a multimodal empathetic framework that introduces a confidence-based adaptive reasoning mechanism. It dynamically navigates between an efficient fast path and a deep slow reasoning chain, balancing both cognitive depth and computational cost.
- We design a condensed-temporal dialogue encoder featuring text-guided cross-modal interaction. This architecture

compresses noisy, asynchronous historical streams into a coherent conversational trajectory, alleviating the multimodal token burden.

- Extensive experiments on the large-scale AvaMERG benchmark demonstrate that CoRE achieves superior performance. It consistently outperforms strong LLM-based baselines in emotion recognition, semantic alignment and human-evaluated empathy.

2 Related Work

2.1 Empathetic Response Generation

Empathetic dialogue generation is grounded in affective resonance and cognitive perspective [9, 25, 31]. Early methods focused on injecting predicted emotional priors into responses [25, 31]. Subsequent research introduced cognitive reasoning, utilizing commonsense knowledge, emotion-cause correlations, and Chain-of-Thought (CoT) templates to stabilize inference [4, 34, 37, 54, 56]. Frameworks like MISC [43] and others [2, 11, 15, 21, 34] demonstrate that effective empathy requires higher-level cognitive signals beyond surface sentiment. Ultimately, these mature text-based paradigms provide reasoning foundations for current multimodal extensions [3, 17, 24].

2.2 Multimodal Empathetic Response Generation

Multimodal Empathetic Response Generation (MERG) extends textual reasoning by integrating visual and acoustic channels to capture richer affective cues [24, 32, 51]. Following early benchmarks [53, 58], recent work introduced the AvaMERG dataset [51] for high-quality text-speech-vision dialogues. Modern frameworks leverage Multimodal Large Language Models (MLLMs) to perform sequential “Chain-of-Empathy” reasoning [16, 24, 49–51]. Recent advancements further incorporate personality traits [45], cognitive therapy strategies [19], and diffusion-based noise restoration [57]. Collectively, these methods transition the field toward cognitively grounded agents capable of authentic multimodal empathy [44].

2.3 Reasoning Redundancy

Reasoning redundancy in Large Reasoning Models (LRMs) refers to the generation of verbose CoT sequences, which increases latency and error accumulation for trivial tasks [1, 6, 38]. To optimize efficiency, researchers employ inference-time pruning [8, 40, 47] or confidence estimation for early termination [12]. Training-based approaches further encourage concise reasoning [28, 30] or distill logical structures into smaller models [14]. Although benchmarks like *OptimalThinkingBench* [1] evaluate these trade-offs, dynamically balancing over- and under-thinking in complex multimodal empathetic contexts remains underexplored, motivating our confidence-driven fast-slow reasoning mechanism.

3 CoRE System

A multimodal dialogue is modeled as a sequence of alternating utterances $D_i = \{U_1, U_2, \dots, U_i\}$, where each utterance U_j consists of visual (V_j), acoustic (A_j), and textual (T_j) signals. The objective

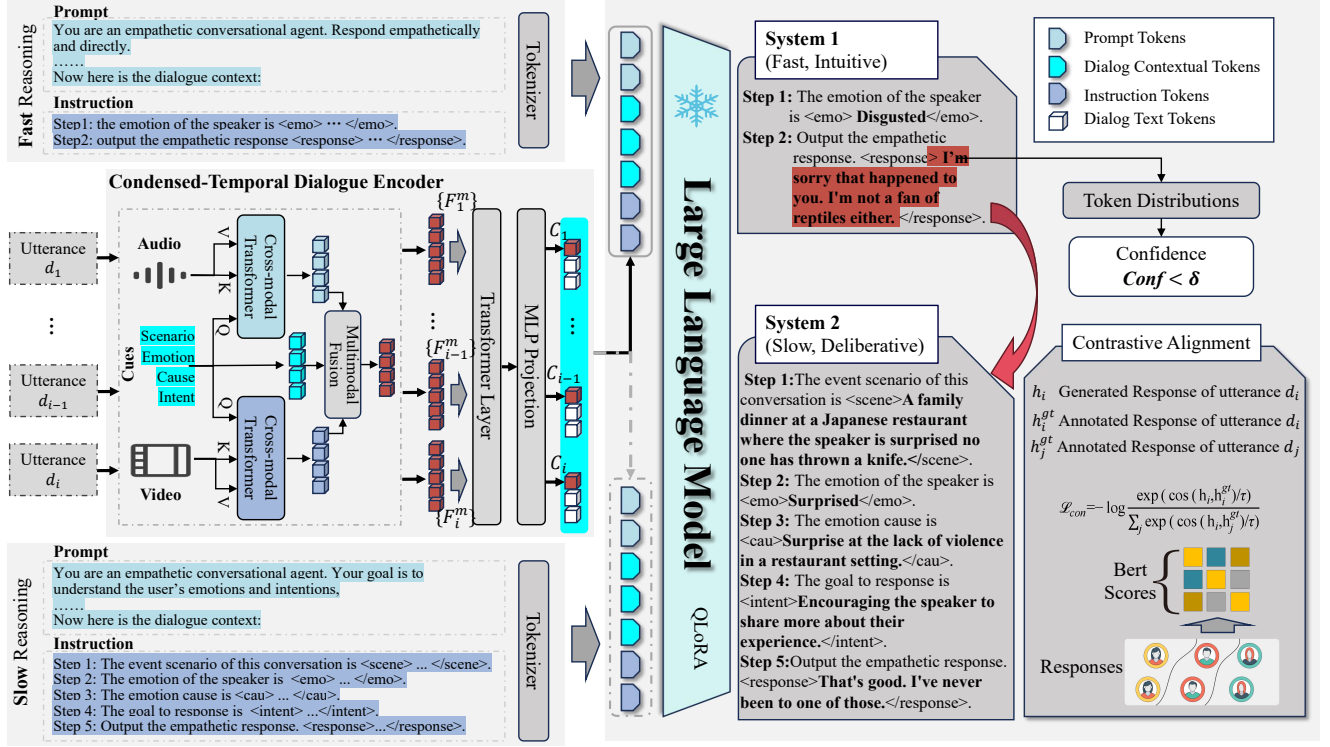


Figure 2: The overall architecture of our proposed CoRE framework. It comprises three main components: (1) Condensed-Temporal Dialogue encoder (CTD), which distills historical multimodal signals into compact non-text temporal features to minimize the LLM token burden; (2) Emotion-aware Empathy Reasoning (ERM), which optimizes the generation process through mixed-mode target instruction supervision and contrastive semantic alignment; and (3) Confidence-driven Fast-slow Reasoning (CFR), which dynamically routes the reasoning path between intuitive generation (System 1) and multi-step analysis (System 2) based on confidence during inference.

of MERG is to perceive the speaker’s latent emotional state E_i and generate a contextually appropriate empathetic response R_i .

Figure 2 presents the overall architecture of CoRE. As an advanced cognitive attribute, empathy involves a multi-stage contemplative process, where listeners synthesize various affective cues before formulating an appropriate response. Inspired by this process, CoRE facilitates empathetic reasoning by injecting multimodal emotional priors and external knowledge into a Chain-of-Thought (CoT) process within a Large Language Model (LLM) to infer the speaker’s latent intentions. Specifically, our framework consists of three components. First, a Condensed-Temporal Dialogue (CTD) encoder distills textual, acoustic, and visual cues via text-guided cross-modal interaction and multi-turn temporal modeling, mapping long-range emotional transitions into a coherent conversational trajectory. During the training phase, an Emotion-aware Empathy Reasoning (ERM) module optimizes the generative space. It employs mixed-mode target instruction supervision alongside a contrastive learning objective to align the latent representations with human-authored empathetic responses to improve semantic and affective consistency. Finally, at inference time, a Confidence-driven Fast-slow Reasoning (CFR) mechanism dynamically routes

the reasoning path based on predictive confidence, striking an balance between cognitive depth and computational efficiency.

3.1 Problem Formulation

An LLM serves as the backbone of our framework, tasked with decoding intricate multimodal cues and orchestrating contextually grounded empathetic actions. Recognizing its robust performance in complex reasoning tasks, we adopt Vicuna [7] as the "brain" to facilitate depth empathy reasoning. The primary objective of the CoRE system is to transform aligned multimodal inputs into structured representations for empathetic response generation. Specifically, after encoding the input multimodal dialogue D , LLM ψ is expected to output the reply text and emotion prediction, empathy reasoning is formulated as:

$$O = \psi(\epsilon^{text}(\text{Prompt}), \epsilon^{dialog}(D)) \quad (1)$$

where ϵ^{text} , ϵ^{dialog} denote the Vicuna (LLaMA-based) language backbone [7] and condensed-temporal dialogue encoder, respectively. O represents the formatted output text result. The multimodal dialogue encoder ϵ^{dialog} consists of audio, vision and text encoders. The slow reasoning prompt, designed for more complex

scenarios, will be described later. Here, we introduce the fast reasoning prompt, which enables efficient inference for straightforward cases:

Prompt = "You are an empathetic conversational agent. Respond empathetically and directly."

Reasoning Instructions. To align the text output with the final reasoning task, we guide the LLM to think through the following instructions to derive the final empathetic responses accurately and interpretable,

Step1: the emotion of the speaker is *<emo>...</emo>*.

Step2: output the empathetic response *<response> ... </response>*.

where the segments enclosed by *<emo>* and *<response>* denote the predicted emotional state and the generated empathetic response, respectively.

Fast Reasoning. The model is trained to generate the entire structured sequence in a unified manner, including both the instructional phrases and the contents within the placeholders, rather than performing isolated slot filling. Consequently, the emotion-aware empathy training process can be formally defined as:

$$O^{fast} = \psi(\epsilon^{text}(\text{Prompt}), \epsilon^{dialog}(D_i), \epsilon^{text}(\text{Instru}^{fast})) \quad (2)$$

where Instru^{fast} denote the reasoning instructions. The fast reasoning path is inspired by human intuitive reasoning. It generates emotion predictions and responses directly from multimodal features without complex reasoning.

3.2 Condensed-Temporal Dialogue Encoder

We employ four reasoning instructions (consistent with slow reasoning): event scenario (T_{scn}), speaker emotion (T_{emo}), emotion cause (T_{cau}), and response goal (T_{goal}). We use a pretrained BERT model to extract their features; let F^{scn} , F^{emo} , F^{cau} , and F^{goal} represent the features of the event scenario, speaker emotion, emotion cause, and response goal, respectively.

$$F^{cues} = [F^{scn}; F^{emo}; F^{cau}; F^{goal}] \quad (3)$$

where $[\cdot]$ denotes feature concatenation, $F^{cues} \in \mathbb{R}^{4 \times d}$.

Instruction-Guided Feature Extraction. For the utterance U_i , F_i^A and F_i^V denote the audio features and visual features, respectively. Specifically, F_i^V captures both fine-grained facial dynamics and global environmental contexts to ensure a holistic affective understanding. To selectively attend to emotion-relevant non-textual cues [20], textual instructions serve as queries over the audio and visual representations within a cross-modal Transformer block:

$$\begin{aligned} F_i^{AC} &= \text{Transformer}(F^{cues}, F_i^A, F_i^A) \\ F_i^{VC} &= \text{Transformer}(F^{cues}, F_i^V, F_i^V) \end{aligned} \quad (4)$$

where $F_i^{AC} \in \mathbb{R}^{4 \times d}$ and $F_i^{VC} \in \mathbb{R}^{4 \times d}$. We fuse multimodal embeddings by concatenation and a linear projection,

$$F_i^{VAC} = \text{MLP}([F_i^{AC}; F^{cues}; F_i^{VC}]) \quad (5)$$

where $F_i^{VAC} \in \mathbb{R}^{4 \times d}$. Based on the above process, six types of features are extracted for utterance U_i : F^{cues} , F_i^A , F_i^V , F_i^{AC} , F_i^{VC} , and F_i^{VAC} . We apply average pooling on the cues dimension,

$$f_i^m = \text{AvgPool}(F_i^m), \quad m \in \{cues, A, V, AC, VC, VAC\} \quad (6)$$

Dialog Contextual Modeling. Taking the dialog $D_i = \{U_1, U_2, \dots, U_i\}$, we can obtain six feature sequences, $fea_i^m = \{f_j^m\}_{j=1}^i$, where

$m \in \{cues, A, V, AC, VC, VAC\}$. We apply a Transformer layer to model the multimodal dialog context,

$$\begin{aligned} H_i^m &= \text{Transformer}(fea_i^m, fea_i^m, fea_i^m) \\ fea_i &= \text{MLP}([H_i^m])_{m \in \{cues, A, V, AC, VC, VAC\}} \end{aligned} \quad (7)$$

where $[\cdot]$ denote feature concatenation, $\text{MLP}()$ denote multilayer perceptron network, $fea_i \in \mathbb{R}^{K \times d}$. We concatenate the six features and apply an MLP to compress it into K embeddings.

For the textual modality, raw utterances T_i are embedded via the Vicuna backbone ϵ^{text} . We concatenate the non-text embedding fea_i and textual feature F_i^{text} ,

$$\begin{aligned} F_i^{text} &= \epsilon^{text}(T_i) \\ C_i &= [fea_i; F_i^{text}] \end{aligned} \quad (8)$$

where $C_i \in \mathbb{R}^{L \times d}$, $L = K + L_u$ and L_u is the text length. For the dialog D_i with i utterances, the dialog embedding can be computed by,

$$\epsilon^{dialog}(D_i) = [C_1; C_2; \dots; C_i] \quad (9)$$

3.3 Confidence-driven Fast-Slow Reasoning

The Fast Reasoning path follows the intuitive "System 1" of the human brain, enabling rapid response generation directly from multimodal inputs. However, this intuitive approach may struggle in emotionally ambiguous scenarios, leading to affectively discordant or superficial outputs.

Slow Reasoning. Inspired by the dual-process theory in cognitive science, we introduce a Slow Reasoning mechanism to complement the intuitive generation process. To elicit step-by-step reasoning from the Large Language Model (LLM), we adopt the cognitive prompting strategy introduced by [51]:

Prompt = "You are an empathetic conversational agent. Your goal is to understand the user's emotions and intentions, and respond or comfort them with appropriate language that helps them feel understood and cared for. Avoid rushing into your response; instead, carefully engage in a step-by-step, in-depth analysis before providing an answer."

Instead of directly generating a direct reply, this prompt guides the LLM to explicitly infer latent cognitive states. The structured reasoning chain is formalized as follows:

Step 1: the event scenario of this conversation is *<scene>...</scene>*.
 Step 2: the emotion of the speaker is *<emo>...</emo>*.
 Step 3: the emotion cause is *<cau>...</cau>*.
 Step 4: the response intention is *<intent>...</intent>*.
 Step 5: output the empathetic response *<response> ... </response>*.

where the placeholders *<scene>*, *<emo>*, *<cau>*, *<intent>*, and *<response>* denote the dialogue scenario, the speaker's emotional state, the underlying emotion cause, the response intention, and the final empathetic reply, respectively. The deliberative reasoning process is formally defined as:

$$O^{slow} = \psi(\epsilon^{text}(\text{Prompt}), \epsilon^{dialog}(D_i), \epsilon^{text}(\text{Instru}^{slow})) \quad (10)$$

where Instru^{slow} denotes the multi-step reasoning instructions.

Confidence-Guided Inference. During inference, the input paradigm is formalized as:

$$O = \psi(\epsilon^{text}(\text{Prompt}), \epsilon^{dialog}(D_i)) \quad (11)$$

Given a multimodal input, the model initially performs Fast Reasoning to generate responses efficiently. To dynamically transition between reasoning modes, we quantify the model’s generation confidence based on token-level predictive entropy over the output distribution. Specifically, for a sequence O^{fast} containing T response tokens, let P_j denote the predicted probability distribution of the j -th token over the vocabulary $\mathcal{V}$. The predictive confidence $Conf$ is formally defined as the negative average Shannon entropy across the sequence:

$$Conf = \frac{1}{T} \sum_{j=1}^T \sum_{v \in \mathcal{V}} P_j(v) \log P_j(v) \quad (12)$$

An empirical decision threshold δ is established at the x -th percentile of the confidence values derived from a validation set. This threshold identifies instances where intuitive (fast) reasoning lacks sufficient certainty:

$$O = \begin{cases} O^{fast}, & Conf > \delta \\ O^{slow}, & Conf \leq \delta \end{cases} \quad (13)$$

where O denotes the final output. When the predictive confidence $Conf$ of Fast Reasoning falls below the threshold δ , the system activates Slow Reasoning to perform a more deliberative analysis. This adaptive switching mechanism ensures that the reasoning depth is adapted with the complexity and ambiguity of the current input.

3.4 Training Objectives

To optimize the proposed CoRE framework, we introduce an **Emotion-aware Empathy Reasoning (ERM)** strategy during the training phase. To equip the model with dual-process capabilities, we construct a joint training corpus comprising Fast and Slow reasoning instances with a mixing ratio of ρ to balance generation efficiency and reasoning depth. During training, the generation follows the paradigm defined in Eq. 2 and Eq. 10. By optimizing the generation loss over these instruction tokens alongside the final response, the model learns to auto-regressively predict the appropriate cognitive path based solely on the input context. This generation process is optimized using the standard negative log-likelihood loss, denoted as $\mathcal{L}_{gen}$.

$$\mathcal{L}_{gen} = - \sum_t \log p(y_t | y_{<t}, O) \quad (14)$$

Contrastive Semantic Alignment. To reduce the semantic gap between generated response and human-annotated response, we extract the hidden state h_i^{LLM} from the last layer. This state and the human-annotated response R_i^{gt} are mapped to the same embedding space.

$$\begin{aligned} h_i &= \text{MLP}(h_i^{LLM}) \in \mathbb{R}^{d_c} \\ h_i^{gt} &= \text{BERT}(R_i^{gt}) \in \mathbb{R}^{d_c} \end{aligned} \quad (15)$$

Treating (h_i, h_i^{gt}) as a positive pair against in-batch negatives h_j^{gt} , the InfoNCE loss is defined as:

$$\mathcal{L}_{con} = - \frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\cos(h_i, h_i^{gt})/\tau)}{\sum_{j=1}^B \exp(\cos(h_i, h_j^{gt})/\tau)} \quad (16)$$

where B is the batch size, $\cos(\cdot, \cdot)$ is cosine similarity, and τ is the temperature.

Emotion Supervision. To enhance the emotional sensitivity of the multimodal temporal representations, we introduce an auxiliary cross-entropy loss on the non-text features fea_i :

$$\mathcal{L}_{emo} = \text{Cross_entropy}(\text{Softmax}(\mathbf{W} \cdot \text{fea}_i + \mathbf{b}), y_i^{emo}) \quad (17)$$

where y_i^{emo} is the ground-truth emotion label, and $\mathbf{W}, \mathbf{b}$ are learnable parameters.

Joint Optimization. The overall training objective is a weighted combination of the generation, contrastive, and emotion losses:

$$\mathcal{L}_{total} = \mathcal{L}_{gen} + \alpha \mathcal{L}_{emo} + \beta \mathcal{L}_{con} \quad (18)$$

where α and β are hyperparameters controlling the contributions of emotion supervision and contrastive regularization, respectively.

4 Experiments

4.1 Experimental Settings

Dataset. We evaluate CoRE on the AvaMERG dataset [51], a large-scale multimodal extension of EmpatheticDialogues [36] featuring synchronized text, speech, and facial videos. It comprises 33,048 dialogues and 152,021 utterances annotated with 32 fine-grained emotion categories. As the official test set lacks annotations for supervised evaluation, we split the official training set into train, validation, and test subsets at a 6:2:2 ratio.

Baselines. We compare CoRE with three categories of representative methods: (1) *Traditional Dialogue Models*: MoEL [25], MIME [31], CEM [37], KEMP [21], CASE [56], and EmpSOA [54]; (2) *LLM-based Models*: DCKS [2], LLaMA-7B [42] with different prompting strategies (+ICL [26], +CoT [4], +CKG [15], +KG^{cpe} [34]), and CFEG [4]; (3) *Multimodal LLMs*: Empatheia [51], E3RG [24], and the Qwen-Omni series [41].

Evaluation Metrics. For *automatic evaluation*, we report Perplexity (PPL), ROUGE-L [23], and BertScore (Precision, Recall, F1) [52] to evaluate generation quality. Emotion recognition is evaluated using Accuracy, Recall, and F1 on both the original 32-class [36] and mapped 7-class taxonomies. For *human evaluation*, three annotators independently rate 100 randomly sampled responses on a 1–5 Likert scale in terms of Coherence, Empathy, Informativeness, and Fluency. Additionally, a pairwise A/B test (Win/Tie/Lose) [55] measures overall preference using majority voting.

Implementation Details. We implement CoRE in PyTorch with DeepSpeed and fine-tune Vicuna-7B using 4-bit QLoRA [10]. The hidden dimension is set to $d = 4096$. Audio and visual inputs are encoded by frozen pretrained encoders into temporal sequences of lengths $T_a = 16$ and $T_v = 32$, respectively. The model is trained on a single RTX 4090 (24GB) GPU with a batch size of 4 for up to 6 epochs using early stopping. The loss weights are $\alpha = 0.3$ and $\beta = 0.5$. The fast and slow reasoning paths are sampled with a mixing ratio of $\rho = 7:3$ during training. During inference (batch size 1), the routing threshold δ is set to the 70th percentile of the

Table 1: Overall performance on empathetic response generation and emotion recognition.

Model	Automatic Generation			Emotion Acc (%)		Human Evaluation				
	PPL↓	F1↑	ROUGE-L↑	7-class↑	32-class↑	Coh↑	Emp↑	Inf↑	Flu↑	Avg↑
<i>Traditional Dialogue Models</i>										
MOEL [25]	24.61	53.37	18.53	70.68	50.47	3.00	3.12	3.03	3.81	3.24
MIME [31]	23.38	54.18	19.05	89.73	59.48	3.43	3.54	3.36	3.99	3.58
CEM [37]	23.68	53.82	19.36	77.79	58.10	3.45	3.51	3.45	4.09	3.63
KEMP [21]	23.65	54.31	19.65	77.51	57.90	3.34	3.48	3.32	3.93	3.53
CASE [56]	22.91	53.90	19.54	73.80	52.73	3.21	3.33	3.18	3.72	3.36
EmpSOA [54]	22.00	55.36	20.87	83.17	63.38	3.49	3.58	3.50	4.07	3.66
<i>LLM-based & Multimodal Models</i>										
DCKS [2]	15.68	56.17	22.45	74.53	53.45	3.77	3.77	3.66	4.16	3.84
CFEG [4]	13.25	59.36	24.68	<u>91.83</u>	<u>75.59</u>	4.33	4.21	4.27	4.52	4.33
LLaMA-7B+ICL [26]	19.35	59.16	24.49	88.90	72.66	4.22	4.09	4.16	4.46	4.23
LLaMA-7B+COT [4]	14.91	<u>59.39</u>	<u>24.92</u>	89.31	73.21	4.34	4.30	<u>4.30</u>	4.51	4.36
LLaMA-7B+CKG [15]	19.55	<u>59.00</u>	<u>24.46</u>	89.01	72.97	4.22	4.17	4.12	4.49	4.25
LLaMA-7B+KG ^{ecpe} [34]	19.66	59.02	24.34	90.15	74.20	4.21	4.06	4.00	4.42	4.17
Empatheia [51]	5.92	58.48	23.33	89.64	71.42	4.29	4.27	4.24	<u>4.56</u>	4.34
E3RG [24]	8.02	55.66	16.60	77.07	45.36	4.28	4.25	4.05	4.40	4.25
Qwen-Omni-Turbo [41]	-	54.54	14.67	77.61	38.45	4.13	4.03	3.94	4.10	4.05
Qwen-Omni-Flash [41]	-	55.32	14.41	82.44	49.55	4.30	4.27	4.02	4.47	4.27
Qwen-Omni-Plus [41]	-	56.34	16.67	81.35	46.73	<u>4.38</u>	4.41	4.24	4.49	<u>4.38</u>
CoRE (Ours)	<u>6.96</u>	59.49	24.96	98.40	78.91	4.43	<u>4.40</u>	4.31	4.57	4.43

validation confidence. Both training and inference prompts include fixed in-context demonstrations for formatting consistency.

4.2 Main Results

Table 1 summarizes the results on empathetic response generation and emotion recognition. Due to space constraints, we report the most representative automatic generation metrics (PPL, BERTScore-F1, ROUGE-L), overall emotion accuracy (7-class and 32-class), and human evaluation results. The remaining metrics, including semantic Precision/Recall and emotion F1 scores, are detailed in Appendix A.

Automatic Evaluation. For *response generation*, traditional dialogue models generally achieve lower semantic similarity, while LLM-based baselines generally improve generation quality. CoRE achieves the best overall performance, including recent multimodal LLMs such as Empatheia, E3RG, and the Qwen-Omni series, achieving the best BERTScore F1 (59.49) and ROUGE-L (24.96). Moreover, CoRE achieves a low perplexity (6.96), indicating fluent and stable generation. These results indicate that integrating multimodal temporal dynamics with confidence-driven fast-slow reasoning improves response quality.

For *emotion recognition*, CoRE achieves the best performance, reaching 98.40% accuracy on the 7-class taxonomy and 78.91% on the more challenging 32-class setting. It outperforms both traditional methods and recent multimodal LLMs, suggesting that temporal compression of visual and acoustic modalities helps capture fine-grained emotional cues. The confusion matrix in Appendix B further shows that most remaining errors occur between semantically similar fine-grained emotion categories.

Human Evaluation. The rightmost columns of Table 1 summarize the human evaluation results. In the Likert-scale assessment, CoRE achieves the highest overall score (4.43) and ranks first on Coherence (4.43), Informativeness (4.31), and Fluency (4.57), while obtaining competitive Empathy performance (4.40), comparable to the strongest baseline. It outperforms traditional models and remains superior against powerful LLM-based baselines indicating that the proposed reasoning strategy produces responses that are perceived as more empathetic and contextually appropriate.

Furthermore, the pairwise A/B test in Table 2 further validates the effectiveness of CoRE. CoRE consistently outperforms traditional empathetic dialogue models, achieving overwhelming win rates (e.g., 92.0% against MOEL and 75.0% against CEM). It also maintains competitive advantages over recent LLM-based baselines. For example, CoRE achieves higher preference rates than Empatheia (38.0% vs. 29.0%), E3RG (40.0% vs. 34.0%), and Qwen-Omni-Flash (40.0% vs. 31.0%). These results indicate that the proposed confidence-driven fast-slow reasoning framework generates responses that are consistently preferred by human evaluators.

4.3 Ablation Study

To evaluate the contribution of modalities, architectural components, and loss hyperparameters (see Appendix C for the sensitivity analysis of α and β), we conduct comprehensive ablation experiments.

Modality Ablation. Table 3 details the impact of different modalities, specifically disentangling the visual stream into facial micro-expressions (E) and global visual contexts (I). Relying solely on text (T) yields the lowest performance (87.44% EmoAcc),

Table 2: Human A/B test results on empathetic dialogues.

Models	Win (%)	Tie (%)	lose (%)
CoRE vs MOEL	92.0	6.0	2.0
CoRE vs MIME	76.0	15.0	9.0
CoRE vs CEM	75.0	15.0	10.0
CoRE vs KEMP	79.0	14.0	7.0
CoRE vs CASE	84.0	13.0	3.0
CoRE vs EmpSOA	77.0	13.0	10.0
CoRE vs DCKS	73.0	17.0	10.0
CoRE vs CFEG	41.0	34.0	25.0
CoRE vs LLaMA-7B+ICL	45.0	28.0	27.0
CoRE vs LLaMA-7B+COT	47.0	30.0	23.0
CoRE vs LLaMA-7B+CKG	51.0	26.0	23.0
CoRE vs LLaMA-7B+ KG^{ecpe}	51.0	28.0	21.0
CoRE vs Empatheia	38.0	33.0	29.0
CoRE vs E3RG	40.0	26.0	34.0
CoRE vs Qwen-Omni-Turbo	48.0	32.0	20.0
CoRE vs Qwen-Omni-Flash	40.0	29.0	31.0
CoRE vs Qwen-Omni-Plus	46.0	15.0	39.0

Table 3: Modality ablation of CoRE. T, A, E, and I denote Text, Audio, facial micro-Expressions, and global Image features, respectively. V indicates the full visual modality (E+I).

Modality	PPL↓	F1↑	EmoAcc↑	AllEmoAcc↑
T	7.62	58.78	87.44	70.23
T + A	7.49	58.88	88.93	71.43
T + E	7.84	59.34	96.08	76.54
T + I	7.87	59.28	96.89	77.05
T + V (E+I)	7.91	58.96	97.28	77.33
T + A + E	7.88	58.47	97.52	77.40
T + A + I	7.78	59.05	97.42	77.37
T + V + A (E+I)	6.96	59.49	98.40	78.91

proving textual cues alone cannot capture complex affective states. Adding acoustic (T+A) or granular visual (T+E, T+I) features consistently enhances emotion perception. Notably, while global features (T+I) capture broader environmental cues (96.89% EmoAcc), micro-expressions (T+E) excel at parsing subtle affective dynamics. Integrating both visual streams (T+V) further improves fine-grained accuracy. Ultimately, the full multimodal fusion (T+V+A) achieves a synergistic breakthrough, peaking across all metrics and drastically reducing perplexity to 6.96. This confirms that macroscopic context, microscopic expressions, and audio signals are mutually reinforcing for empathetic reasoning.

Component Ablation. Table 4 reports the ablation study of CoRE components. Removing the CTD module leads to consistent performance degradation across all metrics, validating the effectiveness of jointly modeling text-guided interaction and temporal dynamics. Removing the Text-Guided Interaction module increases perplexity (from 6.96 to 8.43), highlighting its role in filtering modality noise. Removing Temporal Modeling substantially degrades emotion recognition (EmoAcc drops from 98.40% to 92.16%), demonstrating the importance of capturing historical

Table 4: Ablation study on CoRE architectural components.

Architecture	PPL↓	F1↑	EmoAcc↑	AllEmoAcc↑
CoRE	6.96	59.49	98.40	78.91
w/o CTD	7.96	59.25	91.47	74.51
w/o Text-Guided	8.43	58.84	98.09	77.21
w/o Temporal	8.79	58.91	92.16	71.85
w/o $\mathcal{L}_{con}$	8.78	58.60	97.49	77.06
w/o $\mathcal{L}_{emo}$	8.61	59.41	99.53	78.45

Table 5: Efficiency and performance comparison of different reasoning paradigms. Inference metrics are averaged per sample.

Reasoning Strategy	Efficiency Cost ↓		Performance ↑	
	Time (s)	Tokens	F1	AllEmoAcc
<i>Training Strategies</i>				
Fast-only Training	-	-	59.06	77.98
Slow-only Training	-	-	58.91	76.52
<i>Inference Strategies (under mixed-training)</i>				
Fast-only Inference	7.54	44.87	58.74	77.07
Slow-only Inference	11.86	130.05	58.64	<u>77.74</u>
Random Routing	10.01	88.91	<u>59.00</u>	77.54
CoRE (Adaptive Routing)	<u>8.84</u>	<u>79.78</u>	59.49	78.91

affective dynamics. Removing the contrastive alignment objective degrades both generation quality and emotion recognition, confirming its effectiveness in reducing the semantic gap across modalities. Finally, removing emotion supervision slightly reduces fine-grained emotion recognition while increasing perplexity to 8.61, indicating its contribution to modeling subtle affective states. Further validation of the in-context demonstration strategy is provided in Appendix D.

4.4 Discussion

The proposed CoRE framework relies on a mixed-mode training strategy, a confidence-driven fast-slow reasoning mechanism, and a condensed-temporal dialogue encoder module. In this section, we empirically analyze these CoRE design choices to validate their effectiveness and the resulting efficiency-performance trade-offs.

Effect of Fast-Slow Hyperparameters. We first investigate the fast-slow sampling ratio ρ during training (Figure 3). Extremely unbalanced ratios bias the model toward a single reasoning pattern, while a 5:5 ratio incurs higher computational cost without clear gains. The empirical optimum at $\rho = 7:3$ is consistent with Kahneman’s dual-process theory [18], where intuitive responses dominate and deliberate reasoning is selectively invoked. We further analyze the inference entropy threshold x (Figure 4). The 70th percentile achieves the best overall performance, while CoRE remains robust across a broad threshold range (40%–90%), with only marginal variations in performance. This suggests that routing primarily depends on the relative ranking of uncertainty rather than a finely tuned threshold. Additional routing analyses, including entropy calibration and comparisons with alternative uncertainty

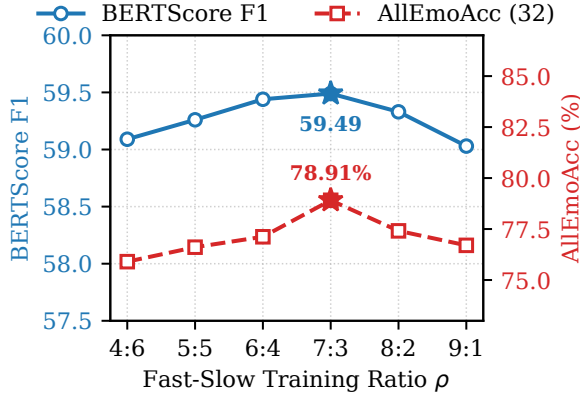


Figure 3: Effect of the fast-slow training ratio ρ on response generation quality (BERTScore F1) and emotion recognition performance (AllEmoAcc).

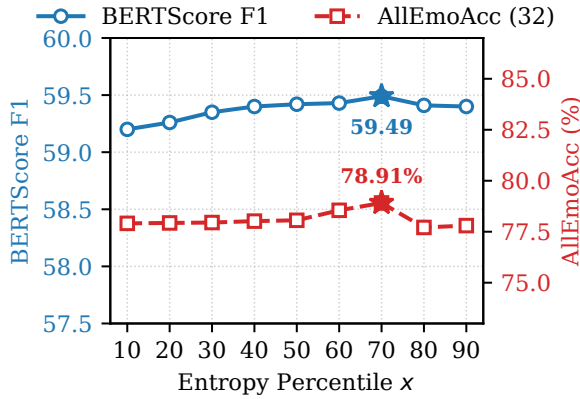


Figure 4: Effect of the entropy percentile threshold x on response generation quality (BERTScore F1) and emotion recognition performance (AllEmoAcc).

measures (Entropy, Energy, MaxProb, and Oracle), are provided in Appendix E.

Efficiency vs. Performance Trade-off. To demonstrate the superiority of our adaptive Fast-Slow mechanism, we compare the computational costs and task performance across different reasoning paradigms in Table 5. For *training strategies*, applying either fast-only or slow-only supervision fails to yield optimal generation quality, proving that jointly learning both paradigms enables the LLM to construct more reliable cognitive representations. During *inference*, enforcing a fast-only strategy consumes the least time and tokens but suffers from performance degradation in ambiguous contexts. Conversely, forcing slow-only reasoning significantly inflates computational overhead (130.05 tokens, 11.86s) without proportional performance gains, often leading to over-thinking on simple queries. Our adaptive routing mechanism effectively bridges this gap. By selectively invoking slow reasoning only when the

Table 6: Effect of the compressed non-textual token length K for non-text features.

Token Length (K)	PPL↓	F1↑	ROUGE-L↑	AllEmoAcc↑
$K = 1$	10.17	59.05	24.53	77.41
$K = 2$	10.43	59.09	24.51	76.74
$K = 4$	6.96	59.49	24.96	78.91
$K = 6$	8.03	58.86	24.23	76.66
$K = 8$	7.59	59.22	24.21	77.94

predictive entropy is high, CoRE achieves state-of-the-art F1 performance (59.49) while maintaining a highly efficient inference speed (a 1.34x speedup compared to the slow-only baseline).

Effect of Non-Textual Token Length. Finally, we evaluate the information bottleneck of our temporal compression by ablating the token length K of the injected non-textual features (F_i^{NT}). As shown in Table 6, extreme compression ($K \in \{1, 2\}$) creates an information bottleneck, forcing the LLM to decode overly dense representations, which spikes perplexity (up to 10.43). The optimal trade-off is achieved at $K = 4$, where CoRE provides sufficient capacity to preserve nuanced emotional signals while maintaining generation fluency. Further extending the sequence ($K \in \{6, 8\}$) introduces redundant visual and acoustic noise, which dilutes the fine-grained performance.

Complementing the empirical evaluations, qualitative observations further validate the practical efficacy of the CoRE framework which is detailed in Appendix F.

5 Limitations

Despite its strong performance, CoRE has several boundary conditions. First, although CoRE is robust to threshold selection, it still relies on an empirically chosen entropy threshold, motivating future learnable routing policies. Second, CoRE still depends on large LLM backbones for multimodal reasoning, limiting deployment on resource-constrained devices. Third, although the CTD encoder effectively compresses temporal information, it may overlook subtle transient visual cues in highly ambiguous scenarios.

6 Conclusion

This paper presents CoRE, a multimodal empathetic reasoning framework that integrates temporal dynamics and dual-process cognition into LLMs. The Condensed-Temporal Dialogue Encoder (CTD) compresses historical audio-visual signals into compact representations. During training, Emotion-aware Empathy Reasoning (ERM) improves affective modeling through mixed-mode supervision and contrastive alignment. During inference, Confidence-driven Fast-slow Reasoning (CFR) dynamically routes queries based on confidence to balance efficiency and reasoning depth. Experiments on AvaMERG demonstrate the effectiveness of CoRE in both response generation and emotion recognition. In future work, we plan to replace the static threshold δ with a reinforcement learning policy that adaptively learns the routing strategy from multimodal feedback.

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References

- [1] Pranjal Aggarwal, Seungone Kim, Jack Lanchantin, Sean Welleck, Jason Weston, Ilia Kulikov, and Swarnadeep Saha. 2025. Optimalthinkingbench: Evaluating over and underthinking in llms. *arXiv preprint arXiv:2508.13141* (2025).
- [2] Hua Cai, Xuli Shen, Qing Xu, Weilin Shen, Xiaomei Wang, Weifeng Ge, Xiaoqing Zheng, and Xiangyang Xue. 2023. Improving empathetic dialogue generation by dynamically infusing commonsense knowledge. In *Findings of the Association for Computational Linguistics: ACL 2023*. 7858–7873.
- [3] Keqi Chen, Wenxin Fu, Qihang Lu, Zekai Sun, Yizhong Geng, Yi Liu, Puyuan Guo, Yingming Gao, and Ya Li. 2025. EMO-Avatar: An LLM-Agent-Orchestrated Framework for Multimodal Emotional Support in Human Animation. In *Proceedings of the 33rd ACM International Conference on Multimedia*. 14014–14020.
- [4] Xinhao Chen, Chong Yang, Man Lan, Li Cai, Yang Chen, Tu Hu, Xinlin Zhuang, and Aimin Zhou. 2024. Cause-aware empathetic response generation via chain-of-thought fine-tuning. *arXiv preprint arXiv:2408.11599* (2024).
- [5] Anqi Cheng, Chengji Wang, Wangda Zhu, Tingting He, and Xiangen Hu. 2026. ABC-Emo: Belief-Catalyzed Emotional Reasoning for Multimodal Emotion Recognition in Conversation. In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26)*. ACM, New York, NY, USA, 1–10. doi:10.1145/3767308.3835906
- [6] Cheng-Han Chiang and Hung-yi Lee. 2024. Over-reasoning and redundant calculation of large language models. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*. 161–169.
- [7] Wei-Lin Chiang, Zhuohan Li, Ziqing Lin, Ying Sheng, Zhanghao Wu, Hao Zhang, Lianmin Zheng, Siyuan Zhuang, Yonghao Zhuang, Joseph E Gonzalez, et al. 2023. Vicuna: An open-source chatbot impressing gpt-4 with 90%* chatgpt quality. See <https://vicuna.lmsys.org> (accessed 14 April 2023) 2, 3 (2023), 6.
- [8] Daewon Choi, Jimin Lee, Jihoon Tack, Woomin Song, Saket Dingliwal, Sai Muralidhar Jayanthi, Bhavana Ganesh, Jinwoo Shin, Aram Galstyan, and Sravan Babu Bodapati. 2025. Think clearly: Improving reasoning via redundant token pruning. *arXiv preprint arXiv:2507.08806* 4 (2025).
- [9] Mark H Davis. 1983. Measuring individual differences in empathy: Evidence for a multidimensional approach. *Journal of personality and social psychology* 44, 1 (1983), 113.
- [10] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. Qlora: Efficient finetuning of quantized llms. *Advances in neural information processing systems* 36 (2023), 10088–10115.
- [11] Fengyi Fu, Lei Zhang, Quan Wang, and Zhendong Mao. 2023. E-CORE: Emotion correlation enhanced empathetic dialogue generation. In *Proceedings of the 2023 conference on empirical methods in natural language processing*. 10568–10586.
- [12] Yichao Fu, Junda Chen, Yonghao Zhuang, Zheyu Fu, Ion Stoica, and Hao Zhang. 2025. Reasoning without self-doubt: More efficient chain-of-thought through certainty probing. In *ICLR 2025 Workshop on Foundation Models in the Wild*.
- [13] Rohit Girdhar, Alaaeldin El-Nouby, Zhuang Liu, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra. 2023. Imagebind: One embedding space to bind them all. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 15180–15190.
- [14] Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Findings of the Association for Computational Linguistics: ACL 2023*. 8003–8017.
- [15] Jena D Hwang, Chandra Bhagavatula, Ronan Le Bras, Jeff Da, Keisuke Sakaguchi, Antoine Bosselut, and Yejin Choi. 2021. (comet-) atomic 2020: On symbolic and neural commonsense knowledge graphs. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 35. 6384–6392.
- [16] Peng Jin, Ryuichi Takanobu, Wancai Zhang, Xiaochun Cao, and Li Yuan. 2024. Chat-univi: Unified visual representation empowers large language models with image and video understanding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 13700–13710.
- [17] Antoine Juqueliere, Ingrid Poncin, and Simon Hazée. 2025. Empathic chatbots: A double-edged sword in customer experiences. *Journal of Business Research* 188 (2025), 115074.
- [18] Daniel Kahneman. 2011. *Thinking, fast and slow*. macmillan.
- [19] Subin Kim, Hoonrae Kim, Jihyun Lee, Yejin Jeon, and Gary Lee. 2025. Mirror: Multimodal cognitive reframing therapy for rolling with resistance. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 14851–14880.
- [20] Jingzhe Li, Chengji Wang, Zhiming Luo, Yuxian Wu, and Xingpeng Jiang. 2024. Modality-dependent sentiments exploring for multi-modal sentiment classification. In *ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 7930–7934.
- [21] Qintong Li, Piji Li, Zhaochun Ren, Pengjie Ren, and Zhumu Chen. 2022. Knowledge bridging for empathetic dialogue generation. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 36. 10993–11001.
- [22] Bill Yuchen Lin, Yicheng Fu, Karina Yang, Faeze Brahman, Shiyu Huang, Chandra Bhagavatula, Prithviraj Ammanabrolu, Yejin Choi, and Xiang Ren. 2023. Swift-sage: A generative agent with fast and slow thinking for complex interactive tasks. *Advances in Neural Information Processing Systems* 36 (2023), 23813–23825.
- [23] Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*. 74–81.
- [24] Ronghao Lin, Shuai Shen, Weipeng Hu, Qiaolin He, Aolin Xiong, Li Huang, Haifeng Hu, and Yap-peng Tan. 2025. E3RG: Building Explicit Emotion-driven Empathetic Response Generation System with Multimodal Large Language Model. In *Proceedings of the 33rd ACM International Conference on Multimedia*. 14006–14013.
- [25] Zhaojiang Lin, Andrea Madotto, Jamin Shin, Peng Xu, and Pascale Fung. 2019. Moel: Mixture of empathetic listeners. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. 121–132.
- [26] Jiachang Liu, Dinghan Shen, Yizhe Zhang, William B Dolan, Lawrence Carin, and Weizhu Chen. 2022. What makes good in-context examples for GPT-3?. In *Proceedings of Deep Learning Inside Out (DeeLIO 2022): The 3rd workshop on knowledge extraction and integration for deep learning architectures*. 100–114.
- [27] Feng Luo, Yu-Neng Chuang, Guanchu Wang, Hoang Anh Duy Le, Shaochen Zhong, Hongyi Liu, Jiayi Yuan, Yang Sui, Vladimir Braverman, Vipin Chaudhary, et al. 2025. Autol2s: Auto long-short reasoning for efficient large language models. *arXiv preprint arXiv:2505.22662* (2025).
- [28] Haotian Luo, Li Shen, Haiying He, Yibo Wang, Shiwei Liu, Wei Li, Naiqiang Tan, Xiaochun Cao, and Dacheng Tao. 2025. O1-pruner: Length-harmonizing fine-tuning for o1-like reasoning pruning. *arXiv preprint arXiv:2501.12570* (2025).
- [29] Mengyuan Ma, Chengji Wang, Sijia Liu, Weizhong Zhao, and Xingpeng Jiang. 2026. BHDD: Facial Behaviors Modeling and Hierarchical Multimodal Fusion for Depression Detection from Vlog Data. *IEEE Transactions on Affective Computing* (2026).
- [30] Xinyin Ma, Guangnian Wan, Runkeng Yu, Gongfan Fang, and Xinchao Wang. 2025. Cot-valve: Length-compressible chain-of-thought tuning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 6025–6035.
- [31] Navonil Majumder, Pengfei Hong, Shanshan Peng, Jiankun Lu, Deepanway Ghosal, Alexander Gelbukh, Rada Mihalcea, and Soujanya Poria. 2020. MIME: MIMicking emotions for empathetic response generation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 8968–8979.
- [32] Ming Nie, Dan Ding, Chunwei Wang, Yuanfan Guo, Jianhua Han, Hang Xu, and Li Zhang. 2024. Slowfocus: Enhancing fine-grained temporal understanding in video llm. *Advances in Neural Information Processing Systems* 37 (2024), 81808–81835.
- [33] Jiabao Pan, Yan Zhang, Chen Zhang, Zuoze Liu, Hongwei Wang, and Haizhou Li. 2024. Dynathink: Fast or slow? a dynamic decision-making framework for large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. 14686–14695.
- [34] Yushan Qian, Bo Wang, Ting-En Lin, Yinhe Zheng, Ying Zhu, Dongming Zhao, Yuexian Hou, Yuchuan Wu, and Yongbin Li. 2023. Empathetic response generation via emotion cause transition graph. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1–5.
- [35] Jingyu Quan, Yoshihiro Miyake, and Takayuki Nozawa. 2025. Incorporating Multimodal Directional Interpersonal Synchrony into Empathetic Response Generation. *Sensors* 25, 2 (2025), 434.
- [36] Hannah Rashkin, Eric Michael Smith, Margaret Li, and Y-Lan Boureau. 2019. Towards empathetic open-domain conversation models: A new benchmark and dataset. In *Proceedings of the 57th annual meeting of the association for computational linguistics*. 5370–5381.
- [37] Sahand Sabour, Chujie Zheng, and Minlie Huang. 2022. Cem: Commonsense-aware empathetic response generation. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 36. 11229–11237.
- [38] Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Na Zou, et al. 2025. Stop overthinking: A survey on efficient reasoning for large language models. *arXiv preprint arXiv:2503.16419* (2025).
- [39] Yiliu Sun, Yanfang Zhang, Zicheng Zhao, Sheng Wan, Dacheng Tao, and Chen Gong. 2025. Fast-Slow-Thinking: Complex Task Solving with Large Language Models. *arXiv preprint arXiv:2504.08690* (2025).
- [40] Keda Tao, Can Qin, Haoxuan You, Yang Sui, and Huan Wang. 2025. Dycoko: Dynamic compression of tokens for fast video large language models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 18992–19001.

- [41] Qwen Team. 2026. Qwen3. 5-omni technical report. *arXiv preprint arXiv:2604.15804* (2026).
- [42] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, et al. 2023. Llama: Open and efficient foundation language models. *arXiv preprint arXiv:2302.13971* (2023).
- [43] Quan Tu, Yanran Li, Jianwei Cui, Bin Wang, Ji-Rong Wen, and Rui Yan. 2022. MISC: A mixed strategy-aware model integrating COMET for emotional support conversation. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: Long papers)*. 308–319.
- [44] Chengji Wang, Wenjing Zhang, Guanyi Chen, Ming Dong, Tingting He, and Xingpeng Jiang. 2026. Divide-and-conquer: Prompt-based distribution learning for multimodal sentiment analysis. *Information Fusion* (2026), 104293.
- [45] Jiaqiang Wu, Xuandong Huang, Zhouan Zhu, and Shangfei Wang. 2025. From traits to empathy: Personality-aware multimodal empathetic response generation. In *Proceedings of the 31st International Conference on Computational Linguistics*. 8925–8938.
- [46] Yuxian Wu, Chengji Wang, Jingzhe Li, Wenjing Zhang, and Xingpeng Jiang. 2024. Uncertainty-Aware Gradient Modulation and Feature Masking for Multimodal Sentiment Analysis. In *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*. Springer, 321–335.
- [47] Heming Xia, Chak Tou Leong, Wenjie Wang, Yongqi Li, and Wenjie Li. 2025. Tokenskip: Controllable chain-of-thought compression in llms. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. 3351–3363.
- [48] Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems* 36 (2023), 11809–11822.
- [49] Jun Zhan, Junqi Dai, Jiaoheng Ye, Yunhua Zhou, Dong Zhang, Zhigeng Liu, Xin Zhang, Ruibin Yuan, Ge Zhang, Linyang Li, et al. 2024. Anygpt: Unified multimodal llm with discrete sequence modeling. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 9637–9662.
- [50] Han Zhang, Hao Fei, Hong Han, Lizi Liao, Erik Cambria, and Min Zhang. 2025. The ACM Multimedia 2025 Grand Challenge of Avatar-based Multimodal Empathetic Conversation. In *Proceedings of the 33rd ACM International Conference on Multimedia*. 14004–14005.
- [51] Han Zhang, Zixiang Meng, Meng Luo, Hong Han, Lizi Liao, Erik Cambria, and Hao Fei. 2025. Towards multimodal empathetic response generation: A rich text-speech-vision avatar-based benchmark. In *Proceedings of the ACM on Web Conference 2025*. 2872–2881.
- [52] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. 2019. Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675* (2019).
- [53] Yiqun Zhang, Fanheng Kong, Peidong Wang, Shuang Sun, SWangLing SWangLing, Shi Feng, Daling Wang, Yifei Zhang, and Kaisong Song. 2024. STICKER-CONV: Generating multimodal empathetic responses from scratch. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 7707–7733.
- [54] Weixiang Zhao, Yanyan Zhao, Xin Lu, and Bing Qin. 2023. Don't lose yourself! empathetic response generation via explicit self-other awareness. In *Findings of the Association for Computational Linguistics: ACL 2023*. 13331–13344.
- [55] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanhao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. 2023. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in neural information processing systems* 36 (2023), 46595–46623.
- [56] Jinfeng Zhou, Chujie Zheng, Bo Wang, Zheng Zhang, and Minlie Huang. 2023. CASE: Aligning coarse-to-fine cognition and affection for empathetic response generation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 8223–8237.
- [57] Xianxun Zhu, Yaoyang Wang, Erik Cambria, Imad Rida, José Santamaría López, Lin Cui, and Rui Wang. 2025. RMER-DT: robust multimodal emotion recognition in conversational contexts based on diffusion and transformers. *Information Fusion* 123 (2025), 103268.
- [58] Zhouan Zhu, Chenguang Li, Jicai Pan, Xin Li, Yufei Xiao, Yanan Chang, Feiyi Zheng, and Shangfei Wang. 2023. Medic: A multimodal empathy dataset in counseling. In *Proceedings of the 31st ACM International Conference on Multimedia*. 6054–6062.